ECONOMICS

ELASTICITY OF DEMAND

Q1 Define price elasticity of demand.

Ans Price elasticity of demand means the degree of responsiveness of demand for a commodity with reference to change in the price of such commodity.

The main demand for measuring price elasticity of demand is :

1. Percentage method :

According to this method, elasticity is measured as the ratio of percentage change in quantity demanded to percentage change in price.

DEGREES OF ELASTICITIES OF DEMAND :

1. Perfectly elastic demand : [ed = infinity]

When there is an infinite demand at a particular price, then the demand for such a commodity is said to be perfectly elastic demand.

In such a case, Ed = infinity and demand curve is a horizontal straight line.

Price	Demand
30	100
30	200
30	300

2. Perfectly Inelastic demand : [Ed = 0] (eg wheat, medicine, etc)

When there is no change in quantity demanded with change in price, then demand for such a commodity is said to be perfectly inelastic demand.

In such a case, Ed = 0 and the demand curve DD is a vertical straight line parallel to Y-axis.

Price	Demand
20	100
30	100
40	100

3. Highly elastic demand : [Ed > 1]

When percentage change in the quantity demanded is more than the percentage change in price, then demand for such a commodity is said to be highly elastic.

In such a case, Ed > 1. The highly elastic demand curve is *flatter.*

Price	Demand
20	100
10	200

4. less elastic demand or inelastic demand : [Ed < 1]

When percentage change in quantity demanded is less than percentage change in price, then demand for such a commodity is said to be less elastic or inelastic.

In such a case, Ed < 1. The inelastic demand curve is *steeper.*

Price	Demand
20	100
10	120

5. Unitary Elastic demand : [Ed = 1]

When percentage change in quantity demanded is equal to the percentage change in price, then demand for such a commodity is said to be unitary elastic.

In such a case, the demand curve is a *rectangular hyperbola.*

In such a case, Ed = 1.

Price	Demand
20	100
10	150

Q1 Arrange the following coefficients of price elasticity of demand in ascending order :

-0.7, -0.3, -1.1, -0.8

Ans -0.3, -0.7, -0.8, -1.1

Q2 The demand for good x is perfectly inelastic for a consumer Saksham. Saksham demands 100 units at a price of Rs 10 per unit. What will be the change in his demand if the price increases to Rs 12?

Ans No change

Q3 Under what condition, the demand curve will be parallel to y-axis?

Ans When demand for a product is *perfectly inelastic,* then the demand curve will be parallel to y-axis.

Q4 under what condition, the demand curve will be parallel to x-axis?

Ans When demand for a product is *perfectly elastic,* then the demand curve will be parallel to x-axis.

FACTORS AFFECTING ELASTICITY OF DEMAND : (imp)

1. Nature of commodity :

Elasticity of demand of a commodity is influenced by its nature.

A commodity may be a necessity, a comfort or a luxury.

When a commodity is a ***necessity*** like food grains, medicines, etc, its demand is generally ***inelastic*** as it is required for human survival.

When a commodity is a ***comfort*** like fan, refrigerator, etc, its demand is generally ***elastic*** as consumer can postpone its consumption.

When a commodity is a ***luxury*** like AC, its demand is generally ***more elastic***.

2. Availability of substitutes :

Demand for a commodity with ***large*** number of substitutes will be ***more elastic.***

For example, a rise in the price of Pepsi encourages buyers to buy coke and vice-versa.

On the other hand, commodities with ***few or no*** substitutes like wheat and salt have ***inelastic demand.***

3. Income Level :

People with ***higher income*** have generally ***inelastic demand*** for any commodity as compared to people with lower income.

4. Number of uses :

If a commodity has ***several uses***, then its demand will be ***highly elastic.*** When price of such a commodity rises, then it is generally put to only more urgent uses and as result, its demand falls.

For example, electricity has several uses.

5. Share in Total expenditure :

Demand for goods like salt, needle, soap, match box, etc have ***inelastic demand*** as consumer spend a ***small*** proportion of their income on such goods.

However, if the proportion of income spend on a commodity is ***large***, then demand for such a commodity will be ***elastic***.

6. Habits :

Commodities which have become habitual necessities (like alcohol, cigarettes, tobacco, etc) for the consumers have ***inelastic demand.***

7. Time period :

Elasticity of demand varies *directly* with the time period.

Demand is generally ***inelastic*** in the ***short period***. It happens because consumers find it difficult to change their habits, in the short period.

However, demand is ***more elastic*** in ***long period*** as it is comparatively easier to shift to other substitutes.

8. Level of price :

Level of price also affects the price elasticity of demand. ***Costly good***s like laptop, AC, etc have ***highly elastic*** demand.

However, ***inexpensive goods*** like needle, matchbox, etc have ***less elastic (inelastic)*** demand.

9. Postponement of consumption :

Commodities like biscuits, softdrinks, etc, whose demand is not urgent, have ***highly elastic*** demand as their consumption can be postponed.

However, commodities with urgent demand like life saving drugs, have ***inelastic demand*** because of their immediate requirement.

Q1 State true or false with reasons.

1. Commodity with diverse uses has generally less elastic demand.

False.

2. Demand is more elastic in the long period than in the short period.

True.

3. A commodity with large number of close substitutes shows high elasticity of demand.

True.

4. When the % increase in demand is more than the % increase in price, then the demand curve is steeper.

False.

5. In case of horizontal straight line demand curve, demand does not change even with change in price.

False

6. In measuring price elasticity, price is a dependent variable and quantity is an independent variable.

False

Q1 Which of the following will have elastic demand

i. matchbox ii. NCERT textbooks iii. medicines iv. air conditioners

Q2 the elasticity of demand for a product will not be higher :

a. When it is considered a necessity by its buyers

b. When more substitutes for the product are available

c. When it has several uses

d. When it is an expensive commodity

Q1 Calculate price elasticity of demand if demand increases from 4 units to 5 units due to fall in price from Rs 10 to Rs8.

Ans

Price	Demand
10	4
8	5

Percentage method

Ed = percentage change in qty demanded/percentage change in price

Percentage change in qty demanded = ¼ x 100 = 25%

Percentage change in price = -2/10 x 100 = - 20%

Ed = percentage change in qty demanded/percentage change in price

Ed = 25/-20 = -5/4 = (-) 1.25

Note : negative sign (-) shows the inverse relationship between price and qty demanded.

Always consider the absolute values.

OR

P = 10 Q = 4 P = -2 Q = 1



Ed = 1/-2 x 10/4 = -10/8 = -5/4 = -1.25

Q2 When price rises from Rs 8 to Rs 10, the demand falls from 5 units to 4 units. Now, elasticity of demand will be :

Q3 When price is Rs 10 per unit, demand for a commodity is 100 units. As the price falls to Rs 8 per unit, demand expands to 150 units. Calculate elasticity of demand.

Q4 When price of sugar is Rs 5 per kg, its demand is 50 kg. when price rises by Rs 5 per kg, its demand falls by 10 kg. Calculate the price elasticity of demand.

Ans

Price	Demand
5	50
5+5 = 10	50 – 10 = 40

Ed = -10/5 x 5/50 = -1/5 = (-)0.2

Q5 The demand for a good falls to 240 units in response to rise in price by Rs 2. If the original demand was 300 units at the price of Rs 20, calculate price elasticity of demand.

Ans

Q5

Price	Demand
10	90
8	100

P = 10, Q = 90

P = -2 Q = 10

Ed = 10/-2 x 10/90 = -100/180 = -10/18 = -5/9 = - 0.55

Q6

Price	Demand
3	3,250
5	1,250

P = 3 Q = 3,250

P = 2 Q = - 2,000

Ed = -2,000/2 x 3/3,250 = - 0.92

Q1 The market demand for a good at Rs 4 per unit is 100 units. Due to increase in price, the market demand falls to 75 units. Find out the new price, if the price elasticity of demand is (-) 1.

Ans

Price	Demand
4	100
?	75

Price = 4

P = 1

New price = 4 + 1 = 5

Q2 When the price of good X is Rs 5, the consumer buys 100 units of good X. At what price would he be willing to purchase 140 units of good X? The price elasticity of demand for good X is 2.

Ans

Q3 A consumer buys 80 units of a good at a price of Rs 5 per unit. Suppose, the price elasticity of demand is (-) 2. At what price will he buy 64 units.

Ans

Price	Demand
5	80
	64

P = 0.5

Δ

New price = 5 + 0.5 = 5.5

Q4 The demand for a good at Rs 10 per unit is 40 units. Price falls by Rs 5. If price elasticity of demand is (-) 3, calculate the new quantity demanded.

Ans 100 units

Q5 Price elasticity of demand is 'unity'. A household buys 25 units at the price of Rs 5. If the price rises by Rs 1, how much quantity of the product will the household buy?

Ans 20 units

Q6 When the price of a commodity falls by Rs 2 per unit, its quantity demanded increases by 10 units. Its price elasticity of demand is (-) 1. Calculate its quantity demanded at the price before change which was Rs 10 per unit.

Ans 50 units

Q7 The initial demand for a commodity is 80 units, the demand falls by 4 units due to rise in price by Rs 10. If price elasticity of demand is 1.5, calculate the price before change in demand.

Ans

Doubts q12,14,19,18

Q12

Price	Demand
10	100
	50

Ed = (-) 2

P = 10, Q = 100, = - 50

Ed = -50/^P x p/Q =

(-) 2 =

Q18

Price	Demand
	60
	55

Q = 60, ^Q = -5, ^P = 1, Ed = (-) 1.5

Q19

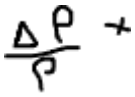
^P = -1, ^Q = 3, Ed = (-) 2, P = 10

Price	Demand
10	?
9	?

ELASTICITY OF DEMAND BY PERCENTAGE METHOD

Percentage change in quantity demanded =

Percentage change in price =



Ed = percentage change in quantity demanded/percentage change in price

Q1 When price of a commodity falls by 80%, the quantity demanded of it increases by 100%. Find out price elasticity of demand.

Ans

Ed = percentage change in quantity demanded/percentage change in price

Ed = 100/80 = 5/4 = 1.25 = (-) 1.25

Q2 A consumer buys 20 units of a good at Rs 10. When its price falls by 10%, its demand rises to 22 units. Find out the price elasticity of demand.

Ans

Percentage change in quantity demanded = ^Q/Q x 100 = 2/20 x 100 = 10%

Ed = 10/10 = 1 = (-) 1 .

Q3 The quantity demanded of a commodity at a price of Rs 8 is 600 units. Its price falls by 25% and the quantity demanded rises by 120 units. Calculate the price elasticity of demand.

Ans

Q4 Calculate the price elasticity of demand for a commodity when its price increases by 25% and quantity demanded falls from 150 units to 120 units.

Percentage change in quantity demanded = 30/150 x 100 = 1/5 x 100 = 20%

Ed = 20/25 = 4/5 = 0.8 = (-) 0.8

MISCELLANEOUS QUESTIONS

Q1 The price elasticity of demand for good X is known to be twice that of good Y. Price of X falls by 5% while that of good Y rises by 5%. What is the percentage change in the quantities demanded of X and Y?

Ans

Good X	Good Y

Elasticity of demand = 2	Let elasticity of demand be 1
% change in price = 5%	% change in price = 5%
Ed = %change in qtd dd/% change in price	
2 = % change in qty dd/5	
% change in qty dd = 10%	***% change in qty dd = 5%***

Q2 The demand for goods X and Y have equal price elasticity. The demand of X rises from 100 units to 250 units due to a 20 percent fall in its price. Calculate the percentage rise in demand of Y, if its price falls by 8 percent.

XY	
7.5	**Ed = 7.5**
	Ed = % change in qty **% change in price**
	7.5 = x **8**
	7.5 x 8 = x
	% change in qty = 60%

Ans 60%

Q3 The price elasticity of demand of good X is half the price elasticity of demand of good Y. A 25% rise in price of good Y reduces its demand from 400 units to 300 units. Calculate the percentage rise in demand of good X when its price falls from Rs 10 to Rs 8.

Ans 10%

ED = 0.5

% change in price

Q4 The percentage change in demand is three times the percentage change in price. If original demand was 30 units at the price of Rs 7 per unit, then calculate the price elasticity of demand, given price increased by 10%. Also calculate the new quantity demanded.

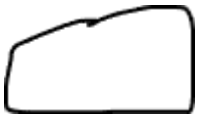
Ans Percentage change in price = 10%

Q = 30 units P = 7

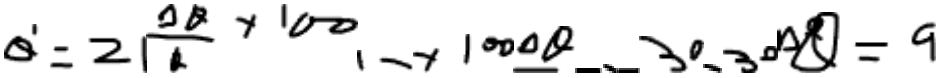
Ed = ?? New qty = ??

% change in demand = 3 x % change in price = 3 x 10 = 30%

Ed = %change in qty dd/%change in price = 30/10 = 3 = (-) 3.



% change in qty dd =



Q5 If ratio of change in quantity (Q) to original quantity (Q) is 0.5 and elasticity of demand is (-) 1.25, calculate the percentage change in price.

Ans 40%

Q6 The demand function of a commodity 'X' is given as : $Q_x = 20 – 2P_x$. Calculate price elasticity of demand when price falls from Rs 5 to Rs 3.

Ans $Q_x = 20 – 2P_x$

When $P_x = 5$, then $Q_x = 20 – 2x5 = 20 – 10 = 10$

When $P_x = 3$, then $Q_x = 20 – 2x3 = 20 – 6 = 14$

Price	Demand
5	10
3	14

Ed = (-) 1

Q41, 44, 64,66,68,75,

Q41

% change in price = 0.2 x 100 = 20%

Ed = % change in qty dd/%change in price

-2 = %change in qty dd/20

%change in qty dd = 40%

Q64

Demand for flowers is more elastic because when price increases by 100% then demand falls by 90%.

However, in toys, when price increases by 100% then demand falls by 50%.

Q66

Price	Demand
10	20
8	24

Ed = 4/-2 x 10/20 = -2 x ½ = (-) 1

Q68

Price	Demand
10	25
12	20

Ed = -5/2 x 10/25 = -1 x 5/5 = (-) 1

Q75

Price	Demand
20	300
?	

%change in qty dd = 10%

Ed = (-) 0.5

Ed = %change in qty dd/%change in price

-0.5 = 10/%change in price

%change in price = 20%

% change in price =

$$\frac{\Delta P}{P} \times 100$$

20 = ^P/20 x 100

20 = 5 x ^P

^P = 4

New price = 20 – 4 = 16

Q1 How does time horizon affect elasticity of demand for a product?

Ans Longer the time horizon, more will be the elasticity of demand for a product.

Q2 The demand of a normal commodity when measured through percentage method is less than unity but more than zero. A fall in its price will result in _____.

a. no change in demand b. decrease in demand

c. increase in demand d. any of the above

Q3 Identify the good whose demand for the good is inelastic.

a. Cigarettes b. Limca c. Burger d. Motor Car

Q4 Under which case demand for a commodity is usually elastic?

a. When a commodity has a less number of close substitutes

b. When a commodity has many uses

c. When a consumer is not in a position to postpone its consumption

d. When a consumer is habitual of using the commodity

Q5 In which type of price elasticity of demand, does the demand curve takes a shape of rectangular hyperbola?

a. Ed = 0 ***b. Ed = 1*** c. Ed = infinity d. Ed < 1

Q6 The demand for a good is said to be inelastic when the percentage change in quantity demanded is _____ the percentage change in the price of the good.

a. equal to b. more than ***c. less than*** d. none

Q7 Ed = 1 means _____.

a. no change in demand due to change in price

b. more proportionate change in demand due to change in price

c. proportionate change in demand is equal to proportionate change in price

d. less proportionate change in demand than price

Q8 What does a steep demand curve indicate?

Ans Inelastic demand

Q9 When demand curve is vertical parallel to y-axis, there will be _____.

a. unitary elastic demand b. perfectly elastic demand

c. perfectly inelastic demand d. relatively elastic demand

HOTS QUESTIONS

Q1 The demand for goods X and Y have equal price elasticity. The demand of X rises from 100 units to 250 units due to a 20 percent fall in its price. Calculate the percentage rise in demand of Y, if its price falls by 8 percent.

Ans 60%

Q2 The price elasticity of demand of good X is half the price elasticity of demand of good Y. A 25% rise in the price of good Y reduces its demand from 400 units to 300 units. Calculate percentage rise in demand of good X when its price falls from Rs 10 to Rs 8 per unit.

Ans 10%

Q3 The percentage change in demand is three times the percentage change in price. If original demand was 30 units at the price of Rs 7 per unit, then calculate the price elasticity of demand, given price increased by 10%. Indicate whether the demand is elastic or not. Also calculate the new quantity demanded.

Ans 3 ; 21 units.

Q4 If ratio of change in quantity (^Q) to original quantity (Q) is 0.5 and elasticity of demand is (-) 1.25 ; calculate the percentage change in price.

Ans 40%

Q6 The demand function of a commodity 'X' is given as : $Q_x = 20 – 2P_x$. Calculate price elasticity of demand when price falls from Rs 5 to Rs 3.

Ans 1

Q7 A consumer spends Rs 2,000 on a good priced at Rs 8 per unit. When price rises by 25%, the consumer continues to spend the same amount on the good. Calculate price easlticity of demand by the percentage method.

Ans 0.80

Q8 A consumer spends Rs 60 on a good priced at Rs 4 per unit. When price falls by 50 percent, the consumer continues to spend Rs 60 on the good. Calculate the price elasticity of demand.

Ans 2

Q9 A consumer spends Rs 1000 on a good priced at Rs 10 per unit. When its price falls by 20 percent, the consumer spends Rs 800 on the good, calculate the price elasticity of demand.

Ans 0

Q10 Price of a commodity falls from Rs 40 to Rs 30 per unit. Quantity demanded initially was 60 units. By how much the quantity will rise if elasticity of demand is established to be unitary?

Ans 15 units

TOTAL EXPENDITURE METHOD

Total expenditure = Price x qty

Qty = TE

—

P

Price = TE

qty

—

Q1 A consumer spends Rs 80 on a commodity when its price is Rs 1 per unit and spends Rs 96 when its price is Rs 2 per unit. Calculate price elasticity of demand.

Ans

P	Q	TE
1	80	80
2	48	96

P = 1 Q = 80

^P = 1 ^Q = 32

Ed = 32 x 1

1. 80

Ed = (-) 0.4

Q2 Calculate price elasticity of demand.

Price	Total expenditure
5	500
6	420

Ans 1.5

PRICE ELASTICITY OF SUPPLY

Price elasticity of supply refers to degree of responsiveness of supply of a commodity with reference to change in the price of such commodity.

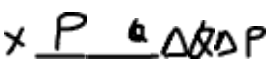
METHOD FOR MEASURING PRICE ELASTICITY OF SUPPLY :

1. Percentage method

Price elasticity of supply (E_s) = % change in quantity supplied/ % change in price

OR

E_s =



Q = change in quantity supplied

Δ

P = change in price

Δ

Eg

Price	**Quantity supplied**
15	**100**
10	**50**

P = 15 Q = 100

P = 5 Q = 50

ΔQ

E_s = 50/5 x 15/100 = 150/100 = 1.5

Since E_s > 1 , therefore the supply is ***highly elastic***

Q1 If price of a commodity falls from Rs 60 per unit to Rs 58 per unit, its supply falls from 400 units to 300 units. Find out its elasticity of supply.

Ans

P	Q
60	400
58	300

P = 60 Q = 400

^P = 2 ^Q = 100

Es = 2/100 x 60/400 = 7.5

Q2 Price of a commodity decreases by Rs 5 per unit and due to this, supply decreases from 100 units to 50 units. Determine price elasticity of supply, if original price was Rs 15 per unit.

Ans 1.5

Q3 The supply for a good is 50 units at the price of Rs 10. When price rises by Rs 5, supply also rises by 50 units. Calculate price elasticity of supply.

Ans 2

Q4 The price of a commodity is Rs 12 per unit and its quantity supplied is 500 units. When its price rises to Rs 15 per unit, its quantity supplied rises to 650 units. Calculate its price elasticity of supply. Is supply elastic?

Ans 1.20 (highly elastic

Q5 The supply for a good rises to 850 units in response to rise in price by Rs 1. If the original supply was 600 units at the price of Rs 4, calculate price elasticity of supply.

Ans 1.67

P	Q
4	600
5	850

Calculation of Elasticity of Supply by percentage method

1. % change in quantity supplied = Q x 100

—Δ

Q

2. % change in price = P. x 100

Δ

P

Q1 When price of a commodity falls by 20%, the quantity supplied decreases by 25%. Find out its price elasticity of supply.

Ans 1.25

Q2 When the price of a good falls from Rs 8 per unit to Rs 6 per unit, its supply falls by 25 units from 125 units. Calculate price elasticity of supply (E_S) by percentage method.

Ans

P	Q
8	125
6	100

P = 8 Q = 125

P = 2 Q = 25



% change in price ^P/p x 100 = 2/8 x 100 = 25%

% change in quantity demanded = ^Q/Q x 100 = 25/125 x 100 20%

E_s = % change in quantity demanded / % change in price = 20/25 = 4/5 = 0.8

Q3 The price of a commodity is Rs 10 per unit and its quantity supplied at this price is 500 units. If its price falls by 10 percent and quantity supplied falls to 400 units, calculate its price elasticity of supply by percentage method.

Ans 2

Q4 Quantity supplied of a commodity increases by 25% when its price rises from Rs 4 to Rs 5. Calculate price elasticity of supply.

Ans 1

When Elasticity of Supply is given

Q1 The coefficient of elasticity of supply of a commodity is 2. A seller supplies 20 units of this commodity at a price of Rs 10 per unit. Calculate the quantity supplied when price rises by Rs 2.

Ans 28 units

Q2 At a price of Rs 8 per unit, the quantity supplied of a commodity is 200 units. Its price elasticity of supply is 1.5. If its price rises to Rs 10 per unit, calculate its quantiy supplied at the new price

Ans 275 units

Q3 The price elasticity of supply is 2.5. At a price of Rs 5 per unit, its quantity supplied is 300 units. Calculate its quantity supplied at a price of Rs 4 per unit.

Ans 150

Q4 The price elasticity of supply is 2. When its price falls from Rs 10 to Rs8, its quantity supplied falls by 500 units. Calculate the quantity supplied at the reduced price.

Ans 750 units

p	Q
10	
8	

P = 10, q = ??, ^p = 2, ^Q = 500 units

Q5 The coefficient of elasticity of supply of a commodity is 3. How much quantity of the commodity will a seller supply when price rises by Rs 1, if he supplies 30 units at Rs 3 per unit?

Ans 60 units

Q6 The ratio of change in price (^P) to original price (P) for a commodity is 0.6. If price elasticity of supply is 1.5, calculate the percentage change in supply.

Ans 90%

Q7 Commodities X and Y have equal price elasticity of supply. The supply of X rises from 400 units to 500 units due to a 20 percent rise in its price. Calculate the percentage fall in supply of Y if its price falls by 8 percent.

Ans

$E_{sx} = E_{sy}$

Commodity x

Q = 400, ^Q = 100

Percentage change in quantity supplied = 100/400 x 100 = 25%

E_{sx} = 25/20 = 5/4 = 1.25

Commodity Y

E_{sy} = 1.25

E_{sy} = percentage change in supply/ percentage change in price

1.25 = percentage change in supply/8

1.25 x 8 = 10% = percentage change in supply

Q The ratio of elasticity of supply of commodities A and B is 1 : 1.5. 20 percent fall in price of A results in a 40 percent fall in its supply. Calculate the percentage increase in supply of B if its price rises from Rs 10 per unit to Rs 11 per unit.

Ans 30%

Calculation of Price Elasticity of Supply (When total revenue (receipts) are given)

Revenue = Price x quantity sold

Price = Rs 1,000

Quantity sold = 10

Revenue = 1000 x 10 = Rs 10,000

Q1 A producer received Rs 6,000 when the price of a commodity was Rs 60 per unit. The receipts increased to Rs 8,400 when price increased by Rs 10. Calculate the price elasticity of supply.

Ans

Price	Quantity	TR
60	100	6,000
70	120	8,400

P = 60 Q = 100

^P = 10 ^Q = 20

Es = 20/10 x 60/100 = 1.2

Supply is highly elastic

Q2 At a price of Rs 5 per unit of commodity A, total revenue is Rs 800. When its price rises by 20 percent, total revenue increases by Rs 400. Calculate its price elasticity of supply.

Ans

Price	Quantity	TR
5	160	800
6	200	1200

P = 5 Q = 160

^P = 1 ^Q = 40

Es = 40/1 x 5/160 = 5/4 = 1.25

Q3 When price of a commodity falls by just 10%, the total revenue of a firm become half of the original total revenue. If at the new price of Rs 45, only 10 units are supplied, calculate original quantity and price elasticity of supply.

Ans

Price	Quantity TR
x	900
45	10 450

Let original price be Rs x.

New price = original price – fall in price

45 = x – 10/100 x

45 = x – x/10

45 = 9x/10

45 x 10/9 = x

x = 50

Price	Quantity	TR
50	18	900
45	10	450

Original quantity supplied = 18 units

P = 50 Q = 18

^P = 5 ^Q = 8

Es = 8/5 x 50/18 = 4.44

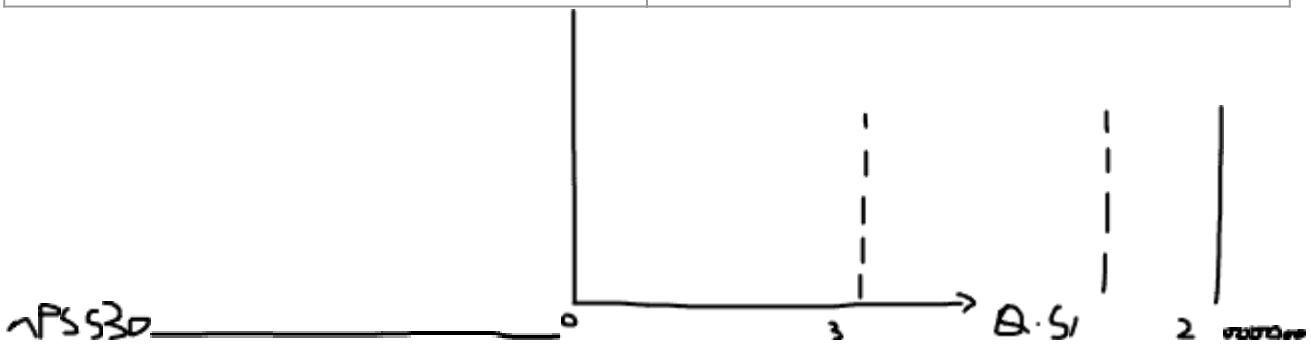
KINDS OF ELASTICITIES OF SUPPLY

1. Perfectly Elastic Supply :

When there is an infinite supply at a particular price, then the supply of such a commodity is said to be perfectly elastic.

In such a case, $\mathbf{E_s}$ **= infinity** and the supply curve is a horizontal straight line parallel to X-axis.

Price	Supply
30	100
30	200
30	300

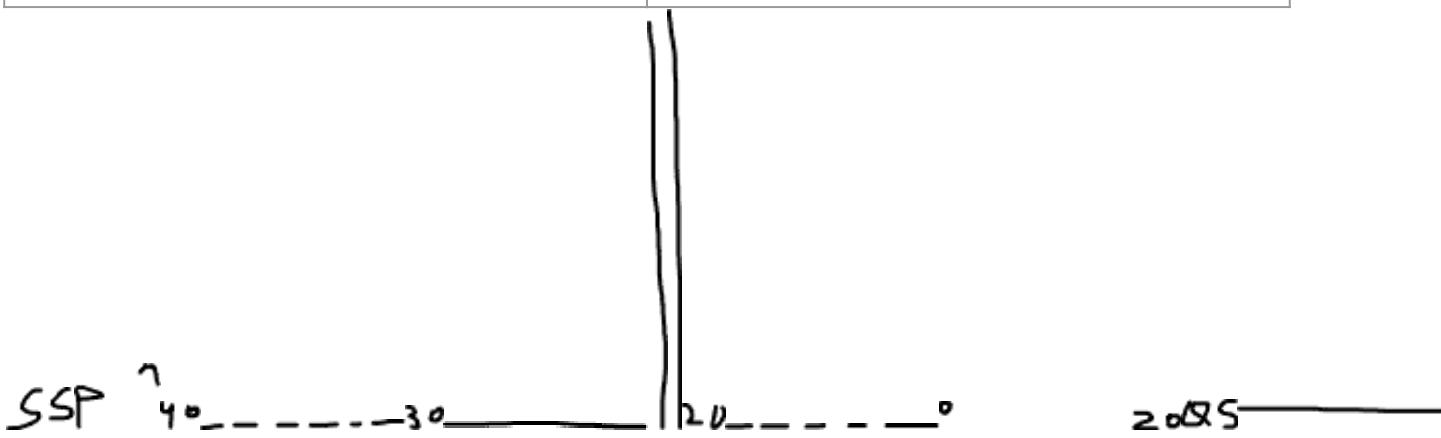


2. Perfectly Inelastic Supply :

When the supply does not change with change in price, then supply for such a commodity is said to be perfectly inelastic.

In such a case, $\mathbf{E_s}$ **= 0** and the supply curve is a vertical straight line parallel to the Y-axis.

Price	Supply
20	20
30	20
40	20

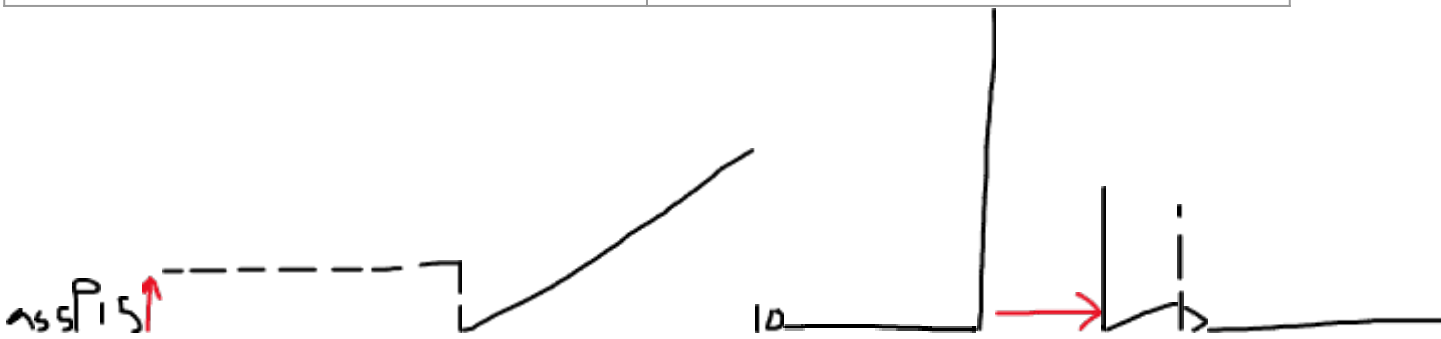


3. Highly Elastic Supply :

When percentage change in quantity supplied is more than the percentage change in price, then supply for such a commodity is said to be highly elastic.

In such a case, $E_s > 1$ and the supply curve has an intercept on the Y-axis.

Price	Supply
10	100
15	200



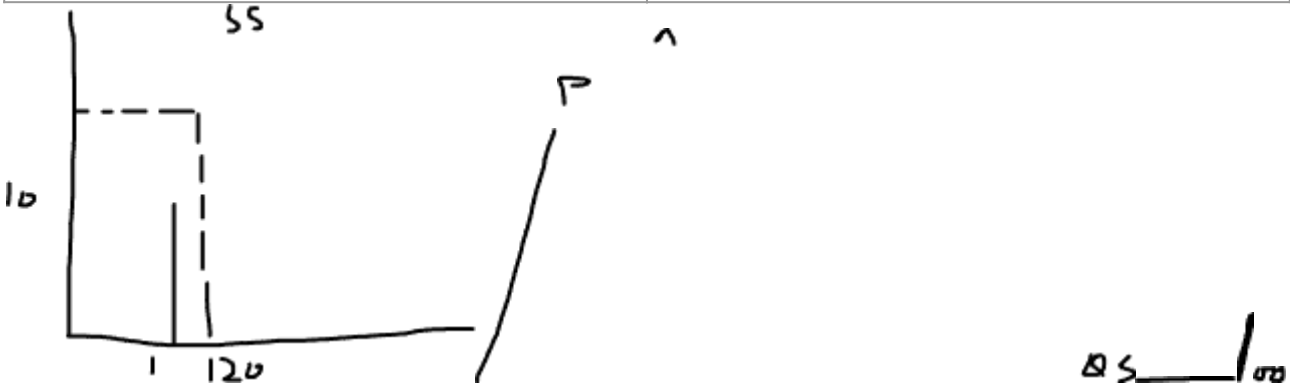
In case of highly elastic supply, the supply curve will be ***flatter.***

4. Less Elastic Supply or Inelastic Supply :

When percentage change in quantity supplied is less than the percentage change in price, then supply for such a commodity is said to be less elastic.

In such a case, $E_s < 1$ and the supply curve has an intercept on the X-axis.

Price	Supply
10	100
15	120

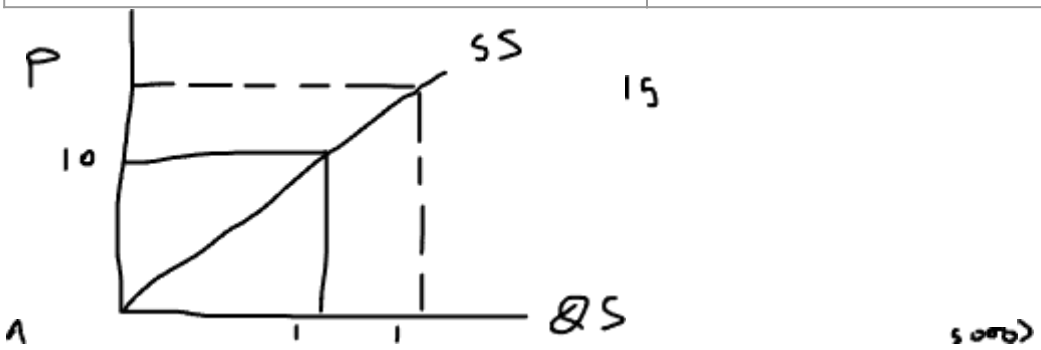


5. Unitary Elastic Supply :

When percentage change in quantity supplied is equal to the percentage change in price, then supply for such a commodity is said to be unitary elastic.

In such a case, E_s = 1 and supply curve is a straight line passing ***through the origin.***

Price	Supply
10	100
15	150



NOTE : All the supply curves which pass through the origin are unitary elastic.

Q5 Are the following statements true or false? Give reasons.

1. Contraction of supply occurs due to change in factors other than price of the given commodity.

Ans **false.** Contraction of supply occurs due to fall in price of the given commodity, keeping other factors constant.

2. Supply is always unitary elastic for all supply curves starting from the origin.

Ans **True.** Any supply curve which passes through the origin has unitary elastic supply.

Q6 At price of Rs 10 per unit, the supply of a good is 80 units. When its price rises to Rs 12 per unit, its supply rises by 20%. Calculate price elasticity of supply.

Ans Es = 1